

Speed, Distance & Time

Practice Questions — Questions 22 to 41

Question 22

A motorbike leaves point A at 1 pm and moves towards point B at a uniform speed. A car leaves point B at 2 pm and moves towards point A at a uniform speed which is double that of the motorbike. They meet at 3:40 pm at a point which is 168 km away from A. What is the distance, in km, between A and B?

- A) 364
 - B) 378
 - C) 380
 - D) 388
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Question 23

In a car race, car A beats car B by 45 km, car B beats car C by 50 km, and car A beats car C by 90 km. The distance (in km) over which the race has been conducted is

- A) 475
 - B) 450
 - C) 500
 - D) 550
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Question 24

A and B are two railway stations 90 km apart. A train leaves A at 9:00 am, heading towards B at a speed of 40 km/hr. Another train leaves B at 10:30 am, heading towards A at a speed of 20 km/hr. The trains meet each other at

- A) 11:45 AM
 - B) 11:20 AM
 - C) 11:00 AM
 - D) 10:45 AM
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Question 25

Two places A and B are 45 kms apart and connected by a straight road. Anil goes from A to B while Sunil goes from B to A. Starting at the same time, they cross each other in exactly 1 hour 30 minutes. If Anil reaches B exactly 1 hour 15 minutes after Sunil reaches A, the speed of Anil, in km per hour, is

- 1) 18
 - 2) 16
 - 3) 14
 - 4) 12
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Question 26

A and B are two points on a straight line. Ram runs from A to B while Rahim runs from B to A. After crossing each other, Ram and Rahim reach their destination in one minute and four minutes, respectively. If they start at the same time, then the ratio of Ram's speed to Rahim's speed is

- A) $\frac{1}{2}$
 - B) $\sqrt{2}$
 - C) 2
 - D) $2\sqrt{2}$
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Question 27

Two circular tracks T1 and T2 of radii 100 m and 20 m, respectively touch at a point A. Starting from A at the same time, Ram and Rahim are walking on track T1 and track T2 at speeds 15 km/hr and 5 km/hr respectively. The number of full rounds that Ram will make before he meets Rahim again for the first time is

- A) 5
 - B) 3
 - C) 2
 - D) 4
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Question 28

Anil, Sunil, and Ravi run along a circular path of length 3 km, starting from the same point at the same time, and going in the clockwise direction. If they run at speeds of 15 km/hr, 10 km/hr, and 8 km/hr, respectively, how much distance in km will Ravi have run when Anil and Sunil meet again for the first time at the starting point?

- A) 4.8
 - B) 4.6
 - C) 5.2
 - D) 4.2
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Question 29

Two cars travel the same distance starting at 10:00 am and 11:00 am, respectively, on the same day. They reach their common destination at the same point of time. If the first car travelled for at least 6 hours, then the highest possible value of the percentage by which the speed of the second car could exceed that of the first car is

- A) 20
 - B) 30
 - C) 25
 - D) 10
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Question 30

A man leaves his home and walks at a speed of 12 km per hour, reaching the railway station 10 minutes after the train had departed. If instead he had walked at a speed of 15 km per hour, he

would have reached the station 10 minutes before the train's departure. The distance (in km) from his home to the railway station is

TITA (Type in the answer)

Question 31

Two ants A and B start from a point P on a circle at the same time, with A moving clock-wise and B moving anti-clockwise. They meet for the first time at 10:00 am when A has covered 60% of the track. If A returns to P at 10:12 am, then B returns to P at

- A) 10:25 AM
 - B) 10:45 AM
 - C) 10:18 AM
 - D) 10:27 AM
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Question 32

Point P lies between points A and B such that the length of BP is thrice that of AP. Car 1 starts from A and moves towards B. Simultaneously, car 2 starts from B and moves towards A. Car 2 reaches P one hour after car 1 reaches P. If the speed of car 2 is half that of car 1, then the time, in minutes, taken by car 1 in reaching P from A is

TITA (Type in the answer)

Question 33

A man travels by a motor boat down a river to his office and back. With the speed of the river unchanged, if he doubles the speed of his motor boat, then his total travel time gets reduced by 75%. The ratio of the original speed of the motor boat to the speed of the river is

- A) $\sqrt{6} : \sqrt{2}$
 - B) $\sqrt{7} : 2$
 - C) $2\sqrt{5} : 3$
 - D) $3 : 2$
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Question 34

In a race of three horses, the first beat the second by 11 metres and the third by 90 metres. If the second beat the third by 80 metres, what was the length, in metres, of the racecourse?

TITA (Type in the answer)

Question 35

Train T leaves station X for station Y at 3 pm. Train S, traveling at three quarters of the speed of T, leaves Y for X at 4 pm. The two trains pass each other at a station Z, where the distance between X and Z is three-fifths of that between X and Y. How many hours does train T take for its journey from X to Y?

TITA (Type in the answer)

Question 36

One can use three different transports which move at 10, 20, and 30 kmph, respectively to reach from A to B. Amal took each mode of transport for $\frac{1}{3}$ rd of his total journey time, while Bimal took each mode of transport for $\frac{1}{3}$ rd of the total distance. The percentage by which Bimal's travel time exceeds Amal's travel time is nearest to

- A) 22
 - B) 20
 - C) 19
 - D) 21
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Question 37

A cyclist leaves A at 10 am and reaches B at 11 am. Starting from 10:01 am, every minute a motorcycle leaves A and moves towards B. Forty-five such motorcycles reach B by 11 am. All motorcycles have the same speed. If the cyclist had doubled his speed, how many motorcycles would have reached B by the time the cyclist reached B?

- A) 22
 - B) 23
 - C) 15
 - D) 20
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Question 38

John jogs on track A at 6 kmph and Mary jogs on track B at 7.5 kmph. The total length of tracks A and B is 325 metres. While John makes 9 rounds of track A, Mary makes 5 rounds of track B. In how many seconds will Mary make one round of track A?

TITA (Type in the answer)

Question 39

The distance from A to B is 60 km. Partha and Narayan start from A at the same time and move towards B. Partha takes four hours more than Narayan to reach B. Moreover, Partha reaches the mid-point of A and B two hours before Narayan reaches B. The speed of Partha, in km per hour, is

- A) 6
 - B) 4
 - C) 3
 - D) 5
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Question 40

Ravi is driving at a speed of 40 km/h on a road. Vijay is 54 meters behind Ravi and driving in the same direction as Ravi. Ashok is driving along the same road from the opposite direction at a speed of 50 km/h and is 225 meters away from Ravi. The speed, in km/h, at which Vijay should drive so that all three cross each other at the same time, is

- A) 58.8

- B) 67.2
 - C) 61.6
 - D) 64.4
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Question 41

A train travelled a certain distance at a uniform speed. Had the speed been 6 km per hour more, it would have needed 4 hours less. Had the speed been 6 km per hour less, it would have needed 6 hours more. The distance, in km, travelled by the train is

- A) 720
 - B) 800
 - C) 780
 - D) 640
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